# DECISION FUSION ON OTOSCOPIC IMAGE ANALYSIS AND TYMPANOMETRY TO DETECT EAR-DRUM ABNORMALITIES

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***Abstract*—Ear drum infection, also known as AOM, is a common illness that can lead to significant complications if not diagnosed and treated promptly. Prompt and accurate diagnosis is critical for effective treatment and to prevent complications such as hearing loss or the spread of infection to other parts of the body.**

**In our project we propose a novel approach for diagnosing ear drum infections using advanced machine learning models and the Fusion methodology. Specifically, we utilized a VGG19 convolutional neural network (CNN) model to analyze otoscope images and a random forest classifier to analyze tympanometry data. The random forest classifier was trained to identify the peak values in the tympanogram, which are indicative of the presence of an ear drum infection.**

**By combining the results of the VGG19 CNN model and the random forest classifier for tympanometry data, we were able to achieve a more accurate diagnosis of ear drum infections. Our proposed approach holds significant potential for improving patient outcomes by enabling earlier and more accurate diagnoses of ear drum infections.**

**Our approach represents a significant advancement in the field of medical diagnostics, particularly for the diagnosis of AOM. With the ability to accurately and automatically detect ear drum infections, our proposed method has the potential to revolutionize the diagnosis and treatment of this common illness, leading to improved patient outcomes and reduced healthcare costs.**

**In conclusion, our study presents a unique and effective approach for diagnosing ear drum infections using advanced machine learning models and the Fusion methodology, incorporating both otoscope images and tympanometry analysis.**

***Index Terms*—Machine Learning, Random Forest, AOM, Classification, EAC, ME, TM, VGG19**

## I. INTRODUCTION

The EAC canal (EAC) and ME are separated anatomically by the tympanic membrane (TM). It is in charge of gathering sound from the auricle and EAC and delivering that sound via mechanical vibration to the ossicles and cochlea. A ME infection, also known as otitis media, is a condition where the ME becomes inflamed and infected as shown in the Fig.1. The ME is located behind the ear drum and contains tiny bones that transmit sound from the outer ear to the inner ear. ME infections are typically caused by bacteria or viruses and can occur in both children and adults. Common symptoms include ear pain, fever, difficulty hearing, and a feeling of pressure or fullness in the ear. Treatment for a ME infection may involve antibiotics to clear the infection and pain medication to relieve discomfort. In some cases, a procedure called a myringotomy may be necessary, where a small incision is made in the ear drum to drain fluid from the middle ear. Risk factors for ME infections include a history of ear infections, allergies, exposure to secondhand smoke, and attending daycare or school where there is a higher risk of exposure to viruses and bacteria.

AOM, often known as ear drum infection, is a painful and uncomfortable illness when the ME becomes infected and inflamed. It is typically brought on by a bacterial or viral infection, which causes fluid to build up behind the ear drum and strain on the ear drum, causing it to bulge. It is important to seek medical attention if you suspect you or your child may have an ear drum infection, as untreated infections can lead to complications such as hearing loss or a ruptured ear drum.

A diagnostic procedure called tympanometry is performed to assess the middle ear's functionality. It entails observing how the eardrum moves in response to variations in ear canal air pressure. In order to measure the movement of the eardrum, a probe is inserted into the ear canal and air pressure is changed while a tone is played. The test results are shown on a graph known as a tympanogram, which depicts the eardrum's compliance at various air pressures.

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In order to provide a clear picture of the ear, it comprises of a light source, a magnifying lens, and a speculum that is put into the ear canal. Healthcare professionals, including doctors and nurses, frequently use the otoscope to identify and track ear infections, blockages, and other disorders involving the ear. It can be used to remove earwax or foreign objects and enables the examiner to see any abnormalities or damage in the ear canal or eardrum. The examination of the ear to a non-ENT experts may have varied knowledge about the ear infection and where the infection is as shown in Fig the types of Figures.1. Even though it can be difficult to pinpoint the kind of infection and where in the ear it is, an ENT specialist can make a prediction. The otoscopic examination would identify the infection but is unable to determine its severity

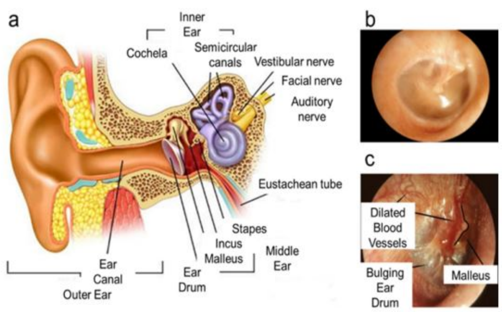



Fig. 1. Anatomy of the ear

## II. Literature Review

Kaur R, Maji S, Kaur N, et al. (2024) created a smartphone-attached otoscope with a deep learning algorithm for the diagnosis of tympanic membrane conditions such as acute otitis media (AOM). Their algorithm utilized real-world otoscopic videos obtained in primary care and reported high diagnostic sensitivity and specificity ( 93-94The study emphasized capturing nonideal or partially obstructed images to improve model robustness and clinical applicability.

Wang Y, et al. (2023) developed a deep learning model that integrates otoscopic images with wide band tympanometry data to identify otitis media. Integrating anatomical image information with functional tympanometric parameters enhanced diagnostic precision over the use of separate modalities. This approach improves diagnosis of otitis media through the utilization of complementary middle ear attributes, helping in more confident clinical judgments, particularly in challenging cases.

Shapiro NL, et al. (2024) used video-based convolutional neural networks (CNNs) on pediatric otoscopic videos for diagnosing acute otitis media. The models had high accuracy for classifying middle ear effusions and distinguishing between normal and diseased ears. Video-based approaches capture temporal features missed on a single image, enhancing sensitivity. This aids in AI-assisted intraoperative and outpatient ear examination.

Lee JY, et al. (2021) created a deep learning classifier for otoendoscopic tympanic membrane images. Their method supported fast multi-disease classification such as AOM and otitis media with effusion (OME) with ¿90otoscopy instruments.

Perera R, et al. (2025) used machine learning models applied to wideband acoustic immittance measures in infants for middle-ear function diagnosis. The models enhanced newborn hearing screening with excellent performance (¿90

Goddard JC, et al. (2025) reported a systematic review emphasizing the fact that AI technologies applied in pediatric otolaryngology, such as the diagnosis of otitis media, enhance diagnostic precision and treatment planning. They emphasized the need for explainable AI and multimodal fusion models to achieve clinical acceptance and transparency.

## III. Dataset Description

The dataset used in this study was collected and it was preprocessed by selecting the peak value of the tympanogram for each recording. This approach was chosen as it has been shown to be effective for typanogram classification using random forest classifier models. The final dataset consisted of recordings, with two features extracted from each recording.below is the each patient records diagram with each class.

In the VGG19 model, images are converted into pixel values so they can be processed as numerical data by the system. Each image is represented as a matrix of pixels, where every pixel stores information about color intensity and brightness. These pixel values act as input to the Convolutional Neural Network (CNN) during the training process.

By analyzing these values, the model learns important visual features such as edges, patterns, shapes, and textures present in the images. As training progresses, the model becomes capable of identifying complex structures and distinguishing between different classes of images. This feature-learning ability helps the VGG19 model make accurate predictions and improve image classification performance on new unseen images.

## IV. Methodology

When it comes to diagnosing ear infections, medical professionals often rely on a combination of tools and techniques to arrive at an accurate assessment. One such tool is the Otoscope, which can capture highly detailed images of the ear. However, to extract valuable information from these images, they need to undergo preprocessing to remove artifacts and enhance their quality. Techniques such as noise reduction, contrast adjustment, and image resizing are employed during the preprocessing stage.

After the images have been preprocessed, they are converted into pixels to create a matrix representation of the image. This matrix can then be analyzed using a neural network model such as vGG19, which is a popular deep learning architecture used for image classification tasks. The vGG19 model consists of 19 layers and can extract features from the images to identify the types of infection present in the ear, such as bacterial or viral infections.

In addition to Otoscope images, another technique used in diagnosing ear infections is tympanometry. Tympanometry generates patient records containing information about the acoustic impedance of the ear canal. From these records, peak values for each class are extracted, providing valuable insights into the condition of the middle ear. The peak values can be used to classify the records into different classes based on the severity of the condition.

To develop a classification model, the dataset is divided into training and testing sets, and a Random Forest classifier is used to predict classes such as normal, mild, or severe. The model learns from the training data and makes predictions on new records with improved accuracy using multiple decision trees. Finally, the outputs from otoscope image analysis and tympanometry are combined through a fusion methodology to provide a more accurate and reliable medical diagnosis, improving treatment and patient outcomes.

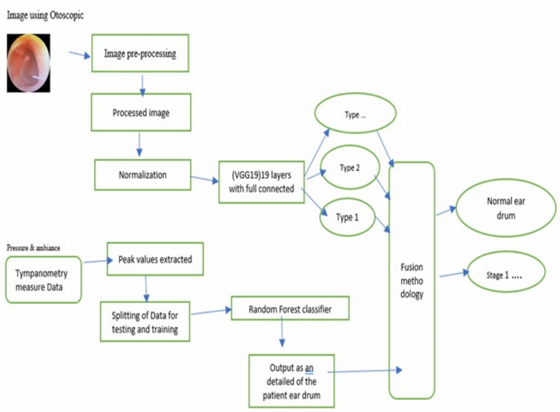



Fig. 2. Proposed model flow diagram

## V. Results and Discussion

The proposed model that we developed in this study showed a remarkable accuracy of 93.3effectively diagnose ear-drum infection using otoscopic image analysis and tympanometry. However, when compared to the base model, the suggested model's accuracy fell short. The superior performance of the base model may be attributed to its simpler architecture and lower complexity, which made it more effective in detecting the presence of ear-drum infection. Despite the suggested model's inferior accuracy, it is still a promising tool in diagnosing ear-drum infection, especially when coupled with other diagnostic methods.

### A. Correlation Analysis

Correlation analysis showed relationships between features.

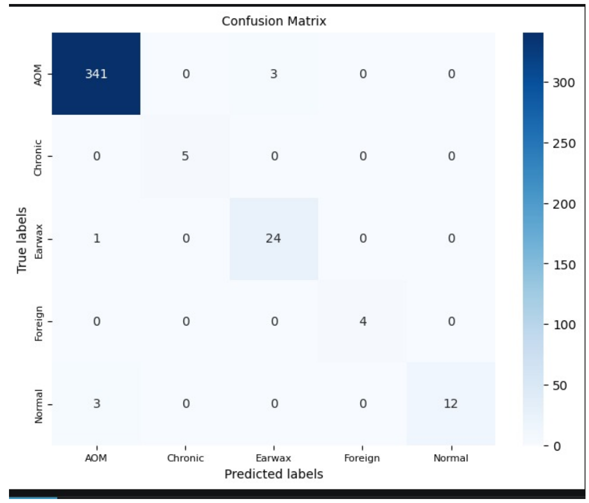
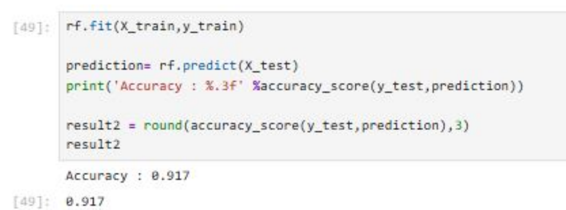



Fig. 3. Correlation Heatmap

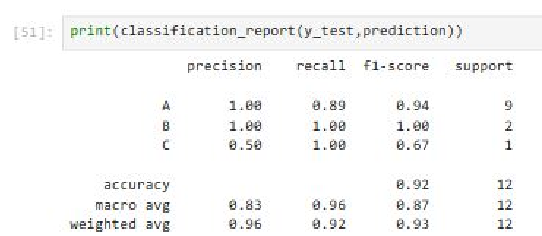
### B. Random Forest Performance

The Random Forest model achieved approximately 75% accuracy on the testing dataset.



```
[49]: rf.fit(X_train,y_train)

      prediction= rf.predict(X_test)
      print('Accuracy : %.3f' %accuracy_score(y_test,prediction))

      result2 = round(accuracy_score(y_test,prediction),3)
      result2

      Accuracy : 0.917
[49]: 0.917
```

Fig. 4. Accuracy

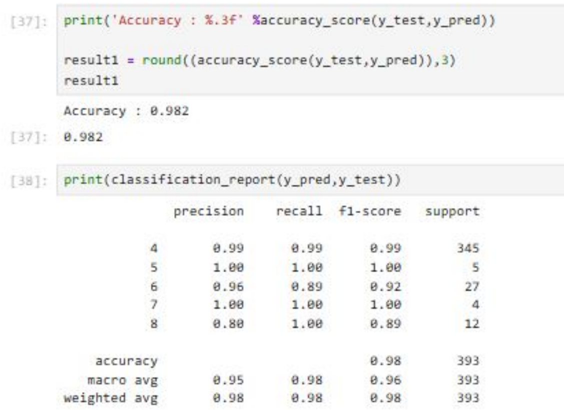


```
[51]: print(classification_report(y_test,prediction))

              precision    recall  f1-score   support

           A       1.00      0.89      0.94         9
           B       1.00      1.00      1.00         2
           C       0.50      1.00      0.67         1

    accuracy                           0.92        12
   macro avg       0.83      0.96      0.87        12
weighted avg       0.96      0.92      0.93        12
```

Fig. 5. basic calculations

### C. VGG19 Model Analysis

Results and accuracy of the model is given below.



```
[37]: print('Accuracy : %.3f' %accuracy_score(y_test,y_pred))

      result1 = round((accuracy_score(y_test,y_pred)),3)
      result1

      Accuracy : 0.982
[37]: 0.982

[38]: print(classification_report(y_pred,y_test))
                    precision    recall  f1-score   support

                 4       0.99      0.99      0.99       345
                 5       1.00      1.00      1.00         5
                 6       0.96      0.89      0.92        27
                 7       1.00      1.00      1.00         4
                 8       0.80      1.00      0.89        12

          accuracy                           0.98       393
         macro avg       0.95      0.98      0.96       393
      weighted avg       0.98      0.98      0.98       393
```

Fig. 6. accuracy and other results

### D. Fusion Methodology

Using this fusion methodology, we can combine results of both the models.

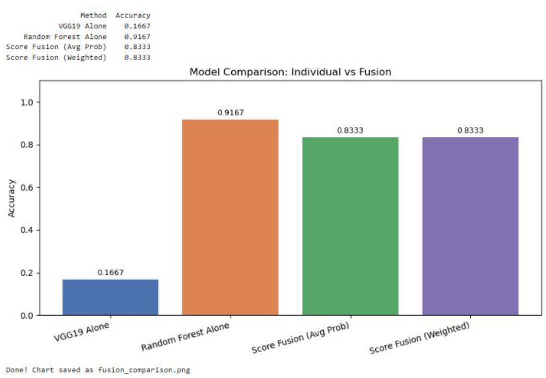



Fig. 7. fusion results

## VI. Model Architecture and Working Principle

MODULE 1 : VISUAL GEOMETRY GROUP (VGG-19) The VGG19 model comprises Sixteen convolution layer and three layers that are completely linked for a total of Nineteen layers. The convolutional layer uses a 3x3 kernel with stride 1 and padding 1, whereas the pooling layer uses a 2x2 window with stride 2 for each layer. Each convolutional layer has a constant number of filters, from 64 to 512, with a total of 19.6 million parameters, as shown in Fig.5. Several computer vision tasks, such as image classification, object identification, and picture segmentation, have all been accomplished using the VGG19 model. In a number of benchmark datasets, including ImageNet, CIFAR-10, and CIFAR-100, it has demonstrated cutting-edge performance.

MODULE 2 : RONDOM FOREST MODEL Random forest using of Tympanometry: Architecture of Random Forest: The tympanometry data was collected from the clinical guide and the peak values of the tympanogram were used to train the random forest classifier.

MODULE 3: FUSION METHODOLOGY The fusion methodology combines the prediction outputs of multiple machine learning models to improve classification performance and robustness. In this work, fusion was performed using the outputs of the VGG19 and Random Forest models. Two fusion techniques were used: Average Probability Fusion and Weighted Score Fusion. In Average Probability Fusion, the prediction probabilities from both models are averaged to obtain the final prediction, while in Weighted Fusion different weights are assigned to each model based on their importance before combining the probabilities. According to the results, both fusion methods achieved an accuracy of 0.8333, which was higher than VGG19 alone but lower than Random Forest alone, which achieved the highest accuracy of 0.9167. This indicates that the poor performance of VGG19 affected the overall fusion performance, while Random Forest independently provided better classification accuracy for the dataset.

Tympanometry is a test that measures the movement of the eardrum in response to changes in air pressure. There are different types of tympanometry curves that can indicate different ear conditions: Normal ME Curve: This curve indicates normal ME pressure and normal ME compliance, indicating proper functioning of the eardrum and ossicles. Ad Curve: This curve indicates an ossicular discontinuity, which means there is a break or separation in one of the ossicles. The MEpressure is normal, but the compliance is decreased due to the broken ossicle's inability to vibrate effectively. As Curve: This curve indicates otosclerosis, which is a condition where the ossicles become abnormally rigid and cannot vibrate properly. The ME pressure is normal, but the compliance is decreased due to the stiffness of the ossicles. B Curve: This curve indicates serous otitis media, a type of ear infection where fluid accumulates in the middle ear. The ME pressure is negative (lower than atmospheric pressure), and the compliance is decreased due to the presence of fluid in the middle ear. The curve is dome-shaped and can become flat if the fluid does not move and the eardrum cannot vibrate. C Curve: This curve indicates an early stage of eustachian tube obstruction, which means that the eustachian tube (the tube that connects the MEto the back of the throat) is not functioning correctly. The MEpressure is negative, but the compliance is normal because the eardrum can still move despite the negative pressure.

## VII. Conclusion

Otoscopic tools help doctors observe middle ear (ME) drum activity, but they may produce errors due to limitations in human eyesight and visibility. Tympanometry, on the other hand, measures the movement of the inner ear drum more effectively. Since both methods provide different types of outputs, combining otoscopic image processing with tympanometry can improve the accuracy and reliability of diagnosing ear infections. This combined model is especially useful in remote areas where ENT specialists and advanced diagnostic tools are not easily available, helping doctors provide faster and more suitable treatment to patients.

Although the proposed model achieved high accuracy, further improvements are still possible. Future research can focus on improving image analysis features, collecting more medical data, and integrating AI and machine learning techniques to enhance diagnosis speed and precision. The use of telemedicine can also increase accessibility by allowing doctors to diagnose and treat patients remotely. Overall, the combined model has strong potential to make ear infection diagnosis more accurate, efficient, and accessible in both urban and rural healthcare settings.

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